

SECTION A

Answer all questions in the spaces provided.

1. Using **outer** electrons only, draw a dot and cross diagram to show the formation of the bonding in magnesium fluoride. [2]

2. In some areas, fluoride ions are added to drinking water.

State **one** benefit of adding fluoride ions to water. [1]

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3. Complete the following definition of relative atomic mass. [1]

The relative atomic mass of an element is the average mass of one atom of the element relative to

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4. Sodium forms only one stable ion. By inserting arrows to represent electrons, complete the electronic structure of this ion. [1]



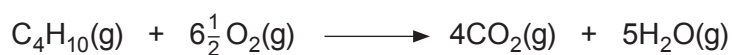
5. Give the oxidation number of vanadium in VOCl_3 . [1]

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6. Phosphoric acid has the formula H_3PO_4 . Write the formula of magnesium phosphate. [1]

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7. Cooking fuel for outdoor camping contains butane, which reacts with oxygen according to the following equation.



If 1.00 mol of butane reacts in this way, calculate the number of **molecules** of carbon dioxide that will be formed. [1]

Molecules of CO_2 =

8. 9.60 g of titanium reacts completely with 3.68 dm^3 of oxygen gas at 298 K and 1 atm to form an oxide.

Calculate the empirical formula of this oxide. [2]

Empirical formula

